Name

Mathematics Stage 5 Paths

Congruence

Book 1

Version: 261008

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Learning Intention ..... **Error! Bookmark not defined.**

Syllabus Content

MA5-GEO-P-01 establishes conditions for congruent triangles and similar triangles and solves problems relating to properties of similar figures and plane shapes

Identify and explain congruence

- Identify figures as congruent figures if translations, reflections and rotations can move one figure exactly on top of another
- Match the sides and angles of 2 congruent polygons
- Indicate 2 polygons are congruent using the symbol (≡) and name the vertices in matching order
- Determine that having equal radii is the condition for 2 circles to be congruent

Develop and use the conditions for congruent triangles

- Establish the 4 congruence tests for triangles (SSS, SAS, AAS and RHS)
- Use the congruence tests to identify a pair of congruent triangles from a selection of 2 or more triangles

Congruence

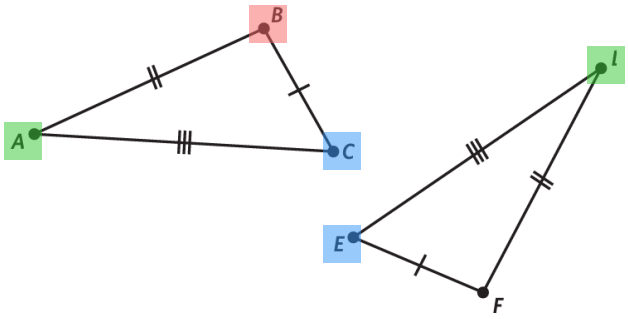
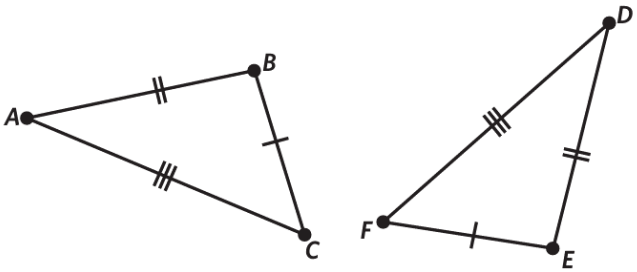
Congruent Shapes

- **Congruent** shapes are identical - they have the same angles, side lengths and area.
- If you **translate** (slide), **reflect** (flip), or **rotate** (spin) a shape, then the image is **congruent** to the original.



- The **corresponding** (or matching) vertices, angles, and sides of two **congruent** shapes are equal.

Example Identify corresponding sides and angles in congruent figures

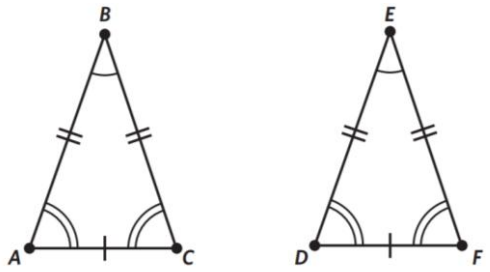
Worked Example	Your Turn
Identify the matching vertices. 	Identify the matching vertices. 

Key Ideas

1. If two figures are ……, then they are exactly identical.
2. If you ……, ……, or …… a shape, then the image is congruent to the original.
3. If two figures are congruent, …… sides and angles are equal.

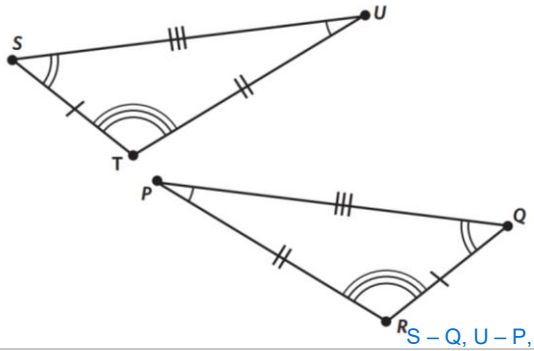
1 Identify the matching vertices.

a



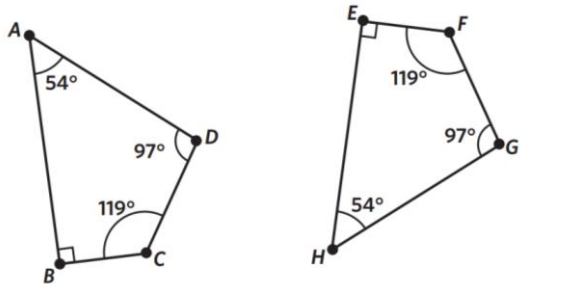
A – D, B – E, C – F

b



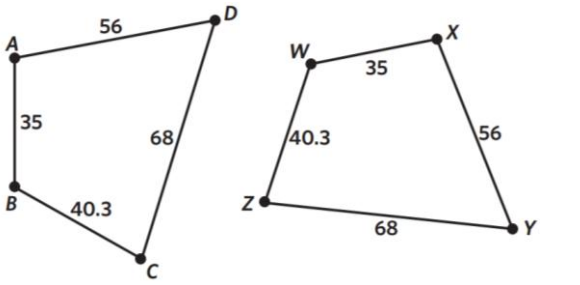
S – Q, U – P, T – R

c



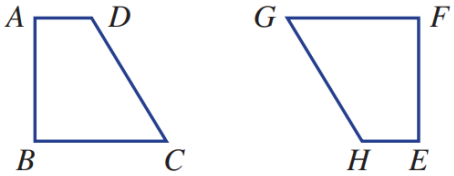
A – H, D – G, C – F, B – E

d



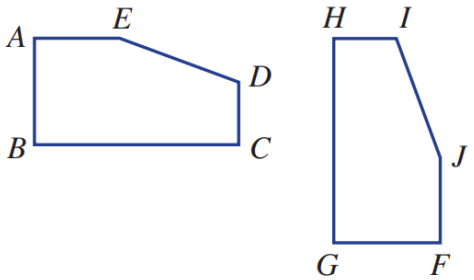
A – X, D – Y, C – Z, B – W

2 These two quadrilaterals are congruent.
Name the feature in EFGH that corresponds to these features in ABCD.



a	A E	**b**	D H	**c**	AD HE
d	CD HG	**e**	∠C ∠G	**f**	∠A ∠E

3 These two pentagons are congruent. Name the object in pentagon FGHIJ that corresponds to these objects in pentagon ABCDE .



a	A F	**b**	D I	**c**	AE FJ
d	CD HI	**e**	∠C ∠H	**f**	∠E ∠J

4 In this diagram △ABC has been translated to give the image triangle △DEF .

a Is △DEF congruent to △ABC?

Yes, rotations are isometric

b Name the vertex on △DEF that corresponds to:

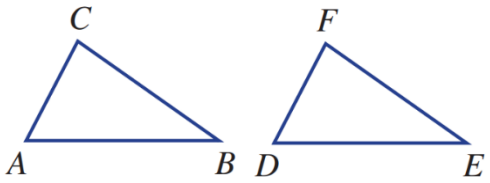
i A ..... D **ii** B ..... E **iii** C ..... F

c Name the side on △DEF that corresponds to:

i AB ..... DE **ii** BC ..... EF **iii** AC ..... DF

d Name the angle in △DEF that corresponds to:

i ∠B ..... ∠E **ii** ∠C ..... ∠F **iii** ∠A ..... ∠D



5 In this diagram △ABC has been reflected to give the image triangle △DEF .

a Is △DEF congruent to △ABC? .....

Yes, reflections are isometric

b Name the vertex on △DEF that corresponds to:

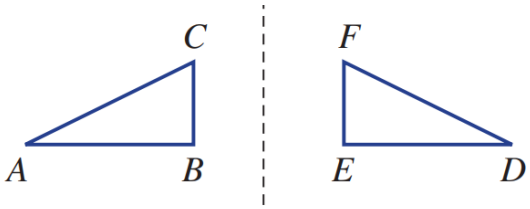
i A ..... D **ii** B ..... E **iii** C ..... F

c Name the side on △DEF that corresponds to:

i AB ..... DE **ii** BC ..... EF **iii** AC ..... DF

d Name the angle in △DEF that corresponds to:

i ∠B ..... ∠E **ii** ∠C ..... ∠F **iii** ∠A ..... ∠D



6 In this diagram △ABC has been rotated to give the image triangle △DEF .

a Is △DEF congruent to △ABC? .....

Yes, rotations are isometric

b Name the vertex on △DEF that corresponds to:

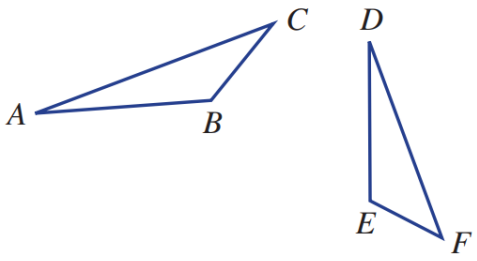
i A ..... D **ii** B ..... E **iii** C ..... F

c Name the side on △DEF that corresponds to:

i AB ..... DE **ii** BC ..... EF **iii** AC ..... DF

d Name the angle in △DEF that corresponds to:

i ∠B ..... ∠E **ii** ∠C ..... ∠F **iii** ∠A ..... ∠D



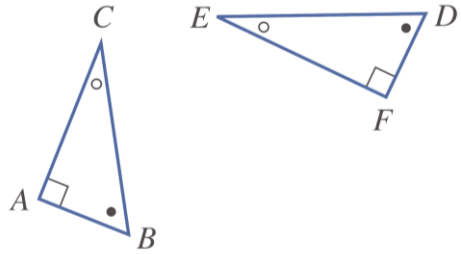
Congruence Statements

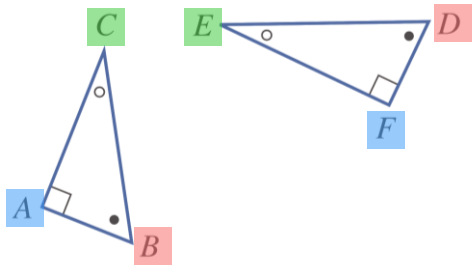
Congruence Statements

- The symbol ≡ indicates congruence.
  1. Identify the corresponding vertices.
  2. Write vertices in corresponding order.

Example Write a congruence statement

Write a congruence statement



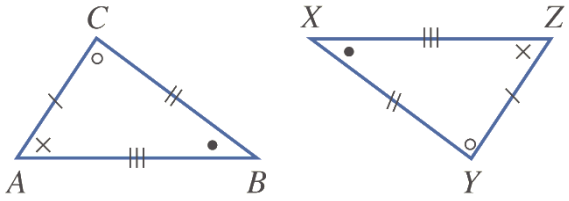
1. 

2. $\Delta ABC \equiv \Delta FDE$

Guided Practice

Write a congruence statement for these congruent shapes.

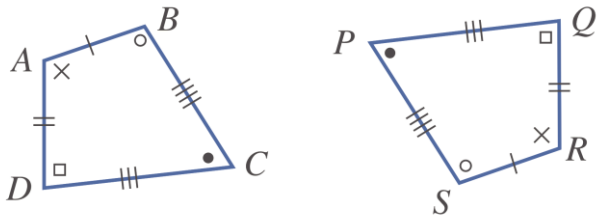
a



Δ ..... ≡ Δ .....

$\Delta ABC \equiv \Delta ZXY$

b



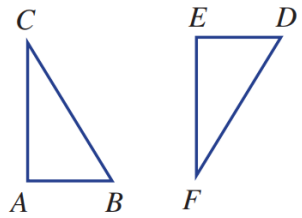
ABCD ≡ $RSPQ$

Key Ideas

1. If ABC is congruent to triangle STU then we write ΔABC …… ΔSTU.
2. When writing congruent statements, the vertices must be in …… order.

1 Write a congruence statement for these shapes.
Remember that the matching vertices should be in the same order.

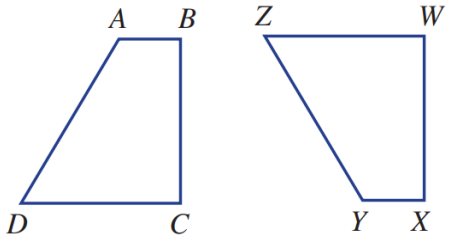
a



Δ ..... ≡ Δ .....

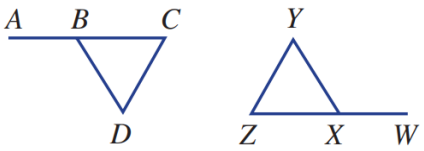
$\Delta ABC \equiv \Delta EDF$

b



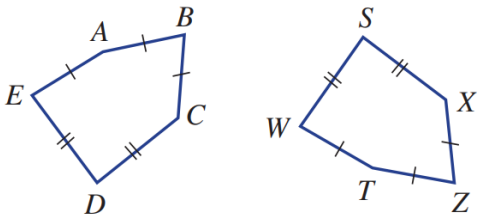
$ABCD \equiv YXWZ$

c



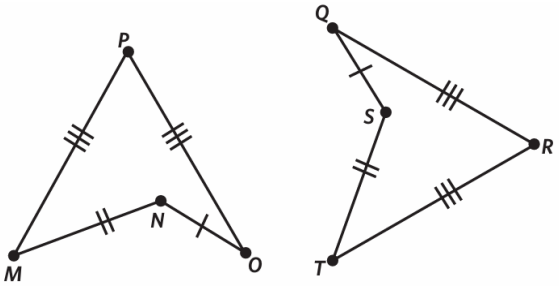
$ABCD \equiv WXZY$

d



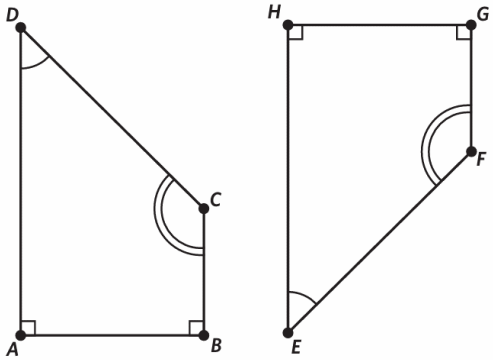
$ABCDE \equiv TZXSW$

e



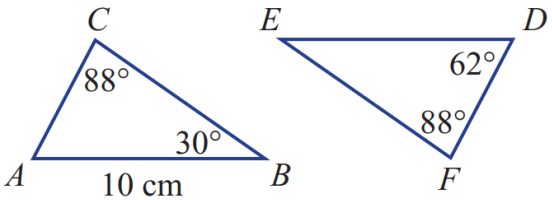
$MNOP \equiv TSQR$

f



$ABCD \equiv HGFE$

2 These triangles are congruent.



a Write a congruence statement:

Δ ..... ≡ Δ .....

$\Delta ABC \equiv \Delta DEF$

b Which side on ΔDEF corresponds to side AB?

ED

c What is the length AB? .....

10

d What is the length DE? .....

10

e Which angle on ΔABC corresponds to ∠E?

$\angle B$

f What is the size of ∠B? .....

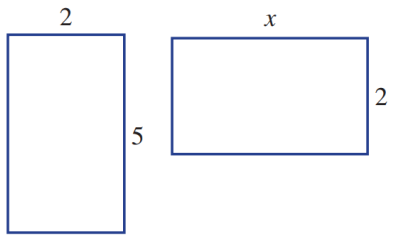
62°

g What is the size of ∠E? .....

30°

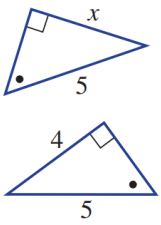
3 These pairs of figures are congruent. Find the value of the pronumeral in each case.

a



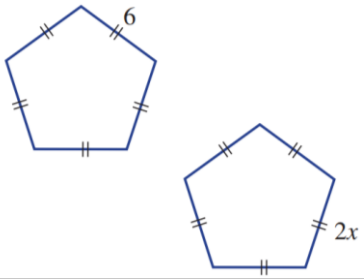
5

b



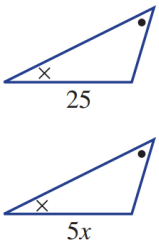
4

c



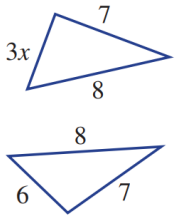
3

d



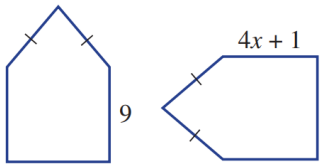
5

e



2

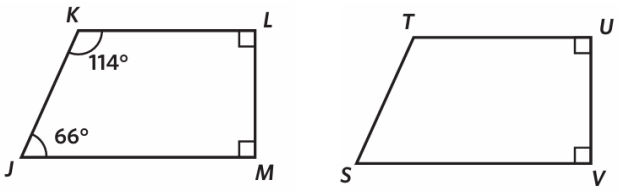
f



2

4 For each shape, write the congruence statement then find the size of the missing angle or side.

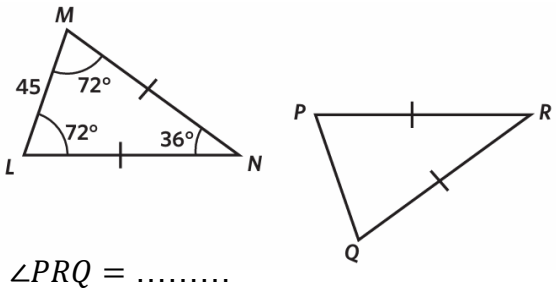
a ..... ≡ .....



$\angle TSV =$ .....

JKLM ≡ STUV, $\angle TSV = 66°$

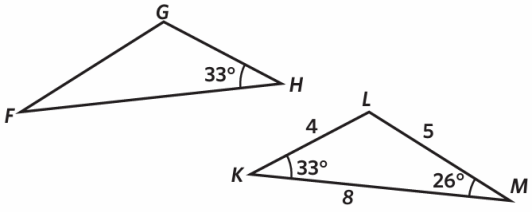
b Δ ..... ≡ Δ .....



$\angle PRQ =$ .....

LMN ≡ QPR, $\angle PRQ = 36°$

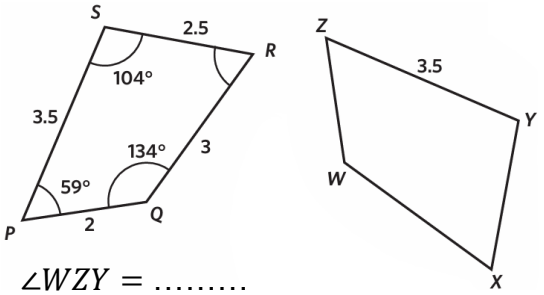
c Δ ..... ≡ Δ .....



$FG =$ .....

FGH ≡ MLN, FG = 5

d ..... ≡ .....

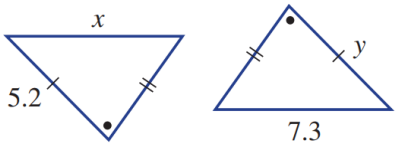


$\angle WZY =$ .....

PQRS ≡ ZWXY, $\angle WXY = 63°$

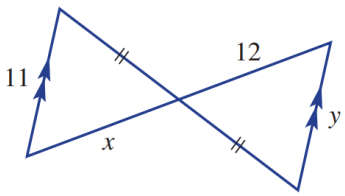
5 Find the value of the pronumerals in these diagrams that include congruent triangles.

a



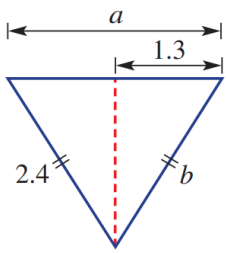
$x = 7.3, y = 5.2$

b



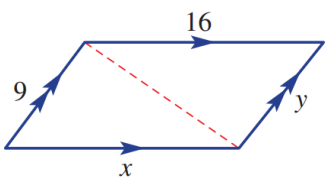
$x = 12, y = 11$

c



$a = 2.6, b = 2.4$

d

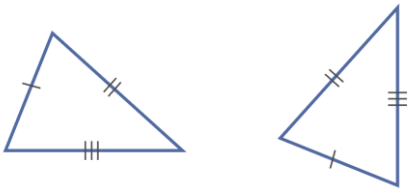


$x = 16, y = 9$

Conditions for Congruent Triangles

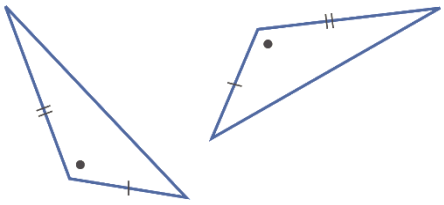
ⓘ Congruent Triangle Tests

SSS



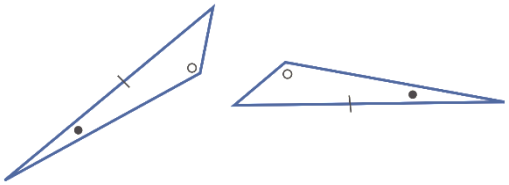
Three pairs of corresponding sides.

SAS



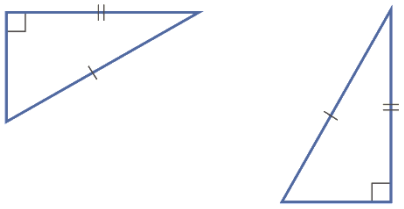
Two pairs of corresponding sides and the included angle equal.

AAS



Two pairs of corresponding angles and one pair of corresponding sides equal.

RHS

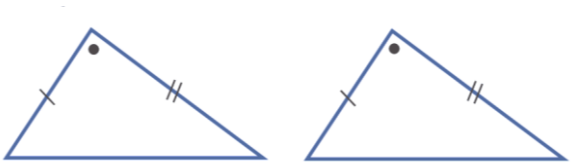


For a right-angled triangle, the hypotenuse and another corresponding side equal.

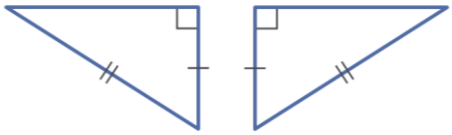
🔍 **Identify** congruent triangles using the four congruence tests

Which of the tests (SSS, SAS, AAS or RHS) prove the congruence of these pairs of triangles?

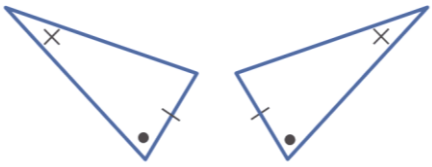
a



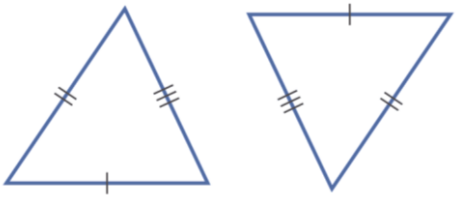
b



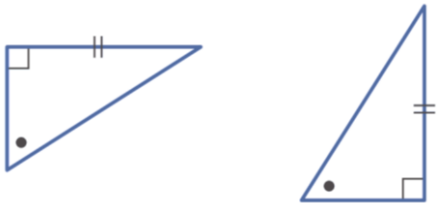
c



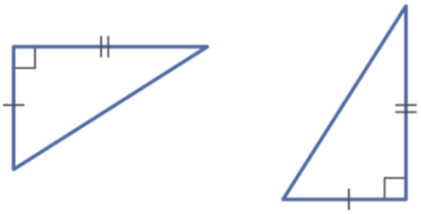
d



e

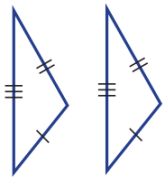


f



1 Pick the congruence test (SSS, SAS, AAS or RHS) for each pair of congruent triangles.

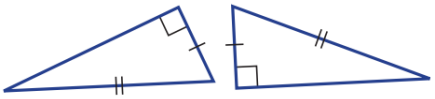
a



❒ SSS ❒ SAS
❒ AAS ❒ RHS

SSS

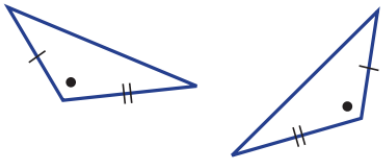
b



❒ SSS ❒ SAS
❒ AAS ❒ RHS

RHS

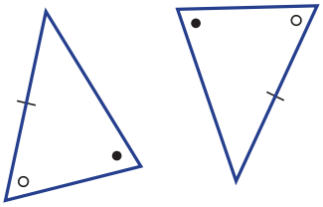
c



❒ SSS ❒ SAS
❒ AAS ❒ RHS

SAS

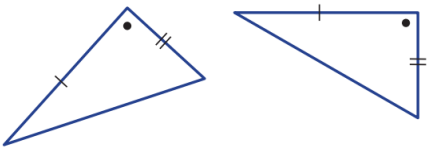
d



❒ SSS ❒ SAS
❒ AAS ❒ RHS

AAS

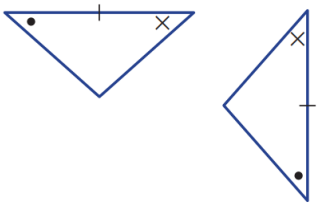
e



❒ SSS ❒ SAS
❒ AAS ❒ RHS

SAS

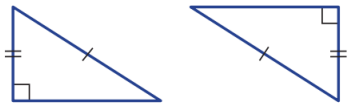
f



❒ SSS ❒ SAS
❒ AAS ❒ RHS

AAS

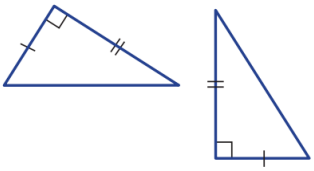
g



❒ SSS ❒ SAS
❒ AAS ❒ RHS

RHS

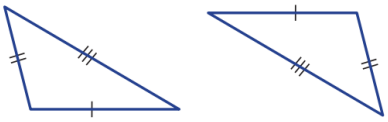
h



❒ SSS ❒ SAS
❒ AAS ❒ RHS

SAS

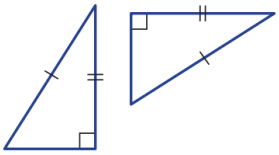
i



❒ SSS ❒ SAS
❒ AAS ❒ RHS

SSS

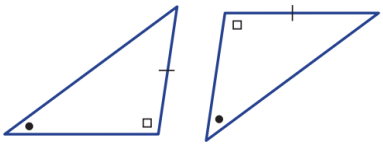
j



❒ SSS ❒ SAS
❒ AAS ❒ RHS

RHS

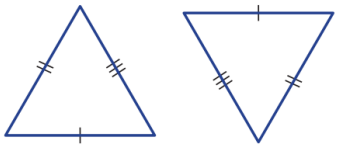
k



❒ SSS ❒ SAS
❒ AAS ❒ RHS

AAS

l

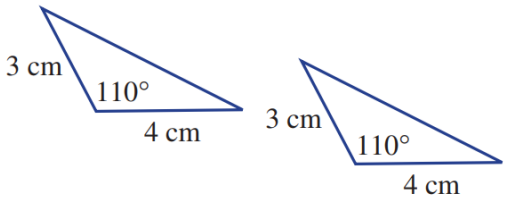


❒ SSS ❒ SAS
❒ AAS ❒ RHS

SSS

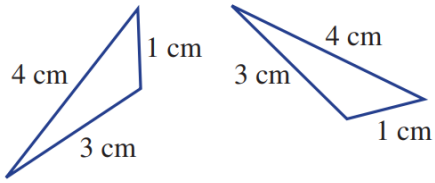
2 Which test (SSS, SAS, AAS or RHS) proves the congruence of these triangles?

a



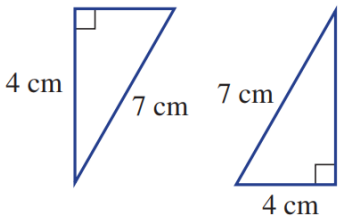
SAS

b



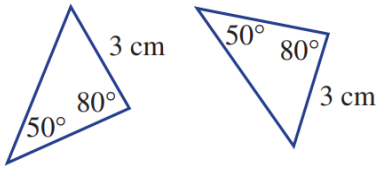
SSS

c



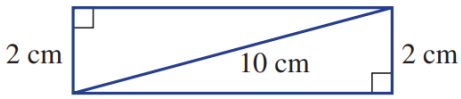
RHS

d



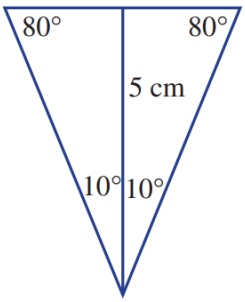
AAS

e



RHS

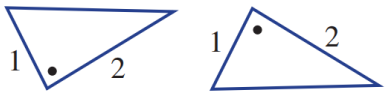
f



AAS

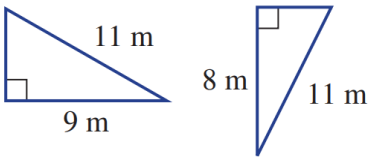
3 Decide if these pairs of triangles are congruent.
If they are, write the test (SSS, SAS, AAS or RHS).

a



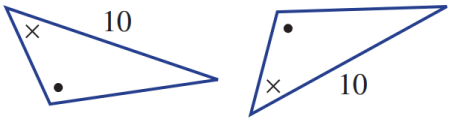
SAS

b



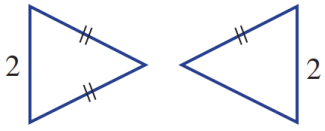
Not congruent

c



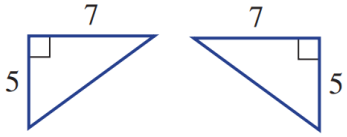
AAS

d



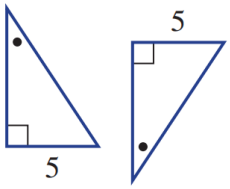
Not congruent

e



SAS

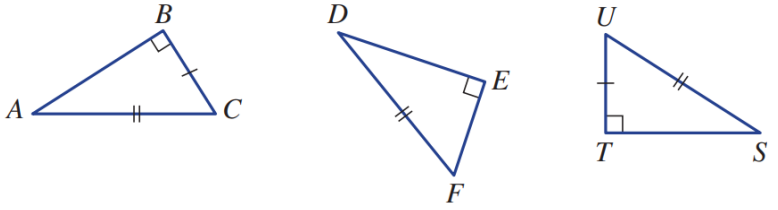
f



AAS

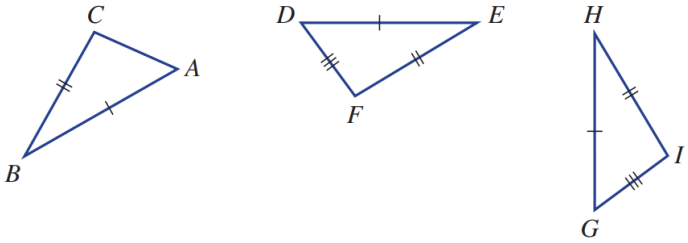
4 For each set of three triangles, choose the two that are congruent. Choose the appropriate test (SSS, SAS, AAS or RHS) and write a congruence statement (e.g. ΔABC ≡ ΔFGH).

a



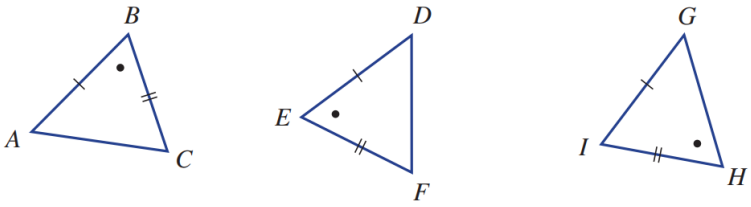
$\Delta ABC \equiv \Delta STU$ (RHS)

b



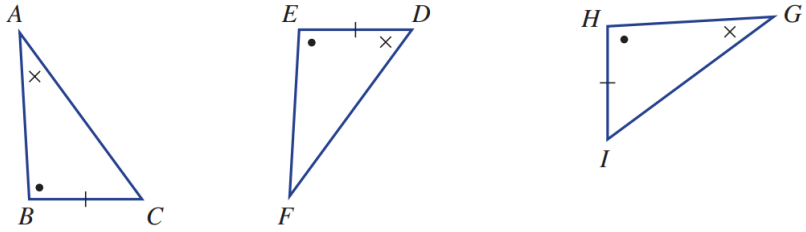
$\Delta DEF \equiv \Delta GHI$ (SSS)

c



$\Delta ABC \equiv \Delta DEF$ (SAS)

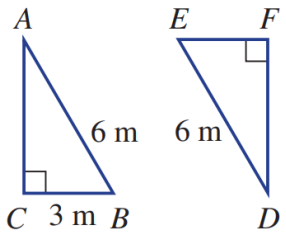
d



$\Delta ABC \equiv \Delta GHI$ (AAS)

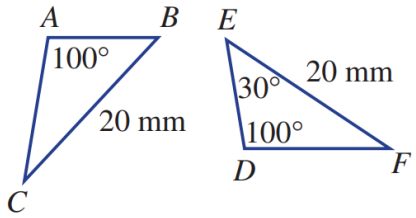
5 Add 1 piece of information to each diagram to make the two triangles congruent, and state the congruence test you are using.

a



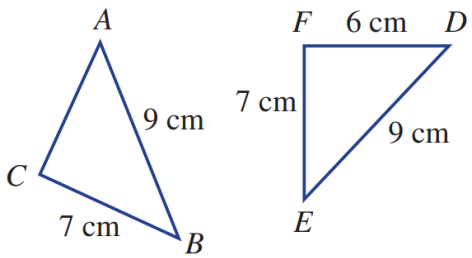
EF = 3 m, RHS

b



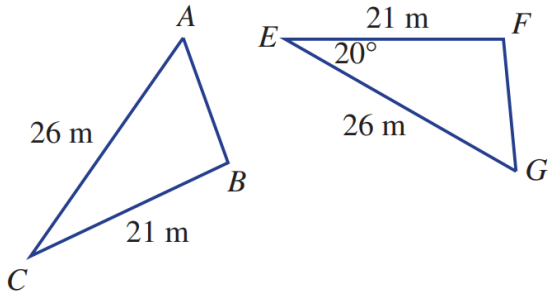
$\angle ABC = 30°$, AAS

c



AC = 6 cm, SSS

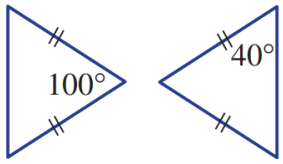
d



$\angle ACB = 20°$, SAS

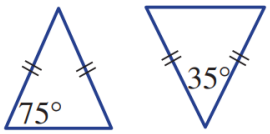
6 Decide if each pair of triangles is congruent.
You first need to use the angle sum of a triangle to help calculate some of the angles.

a



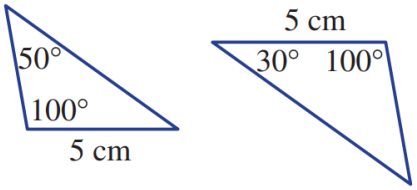
yes

b



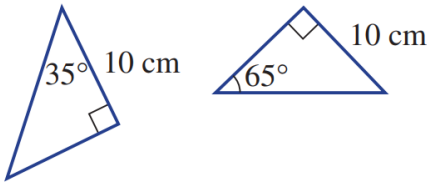
no

c



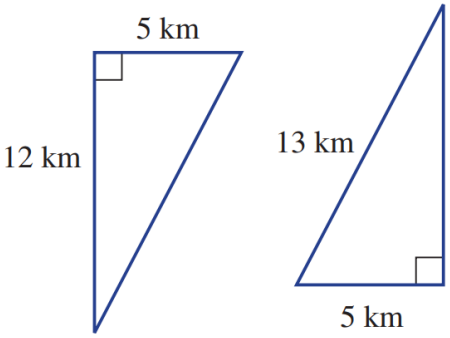
yes

d



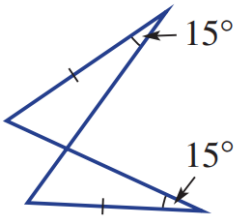
no

7 Two adjoining triangular areas of land have the dimensions shown.
Use Pythagoras' theorem to help decide if the two areas of land are congruent.



yes

8 Consider this diagram including two triangles.



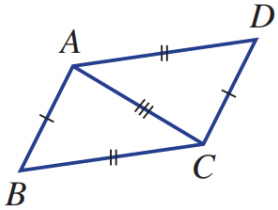
a Explain why there are two pairs of equal matching angles.

There is a pair of equal 15° angles (given)
And there is a pair of equal vertically opposite angles

b Give the reason (SSS, SAS, AAS or RHS) why there are two congruent triangles.

AAS

9 ABCD is a parallelogram.



a Give the reason why ΔABC ≡ ΔCDA.

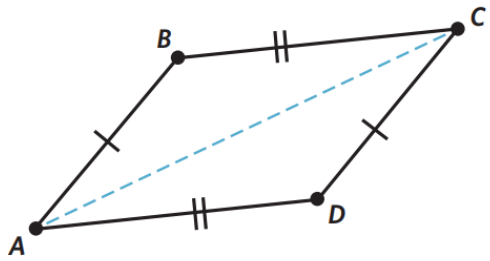
SSS

b What does this say about ∠B and ∠D?

They are the same as corresponding angles in congruent triangles are equal.
This is also a property of a parallelogram (opposite angles equal)

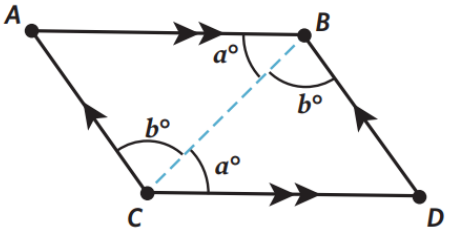
10 Determine a test that could be used to prove these triangles are congruent.

a $\triangle ADC$ and $\triangle ABC$



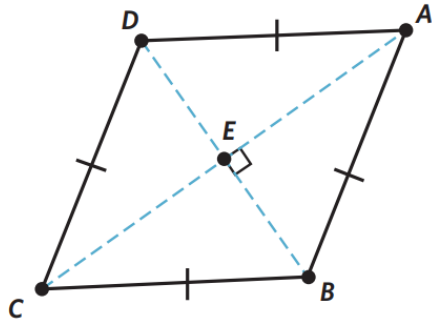
SSS

b $\triangle ABC$ and $\triangle DBC$



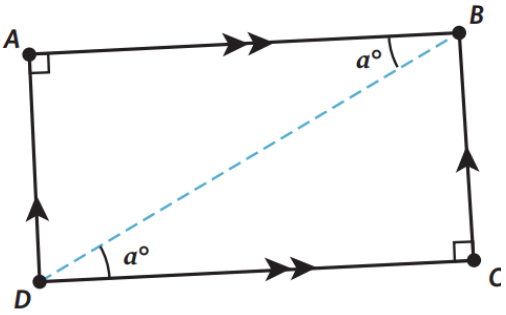
AAS

c $\triangle BEA$ and $\triangle DEA$



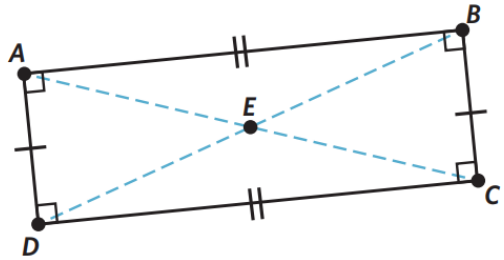
RHS

d $\triangle ABD$ and $\triangle DBC$



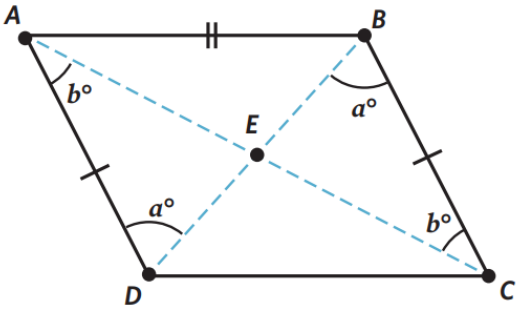
AAS

e $\triangle ABC$ and $\triangle DBC$



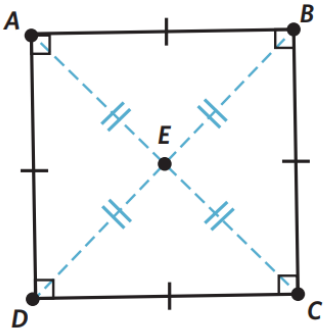
SSS

f $\triangle AED$ and $\triangle BEC$



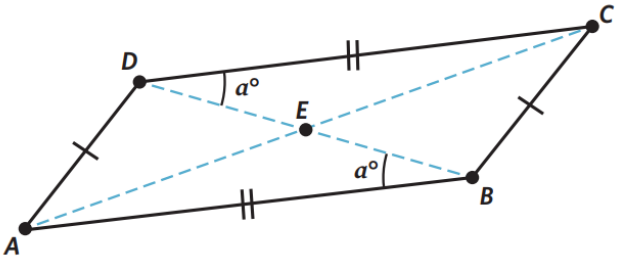
AAS

g $\triangle AEB$, $\triangle AED$, $\triangle DEC$, and $\triangle CEB$



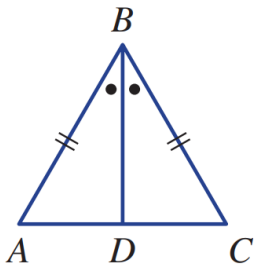
SSS

h $\triangle AEB$ and $\triangle DEC$



AAS

11 Consider this diagram.



a Explain why there are two pairs of matching sides of equal length for the two triangles ΔABD and ΔCBD

AB = BC (given)

BD is a common side

b Give the reason (SSS, SAS, AAS or RHS) why there are two congruent triangles.

SAS

c Write a congruence statement.

$\Delta ABD \equiv \Delta CBD$ (SAS)

d Explain why AC is perpendicular (90°) to DB.

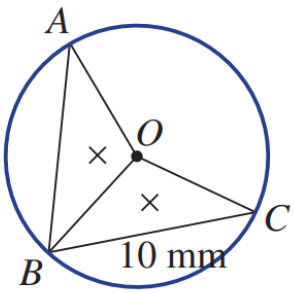
$\angle ADB = \angle CDB$ (corresponding sides on congruent triangles)

But $\angle ADB + \angle CDB = 180°$ (angles on a straight line)

Since the two angles are equal and sum to 180, each angle must be 90°.

So AC is perpendicular to DB

12 In this diagram O is the centre of the circle and ∠AOB = ∠COB.



a Give the reason why △AOB ≡ △COB

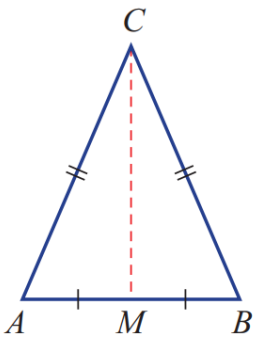
(OA, OB, OC) are all radii of a circle and equal.

SAS

b Hence state the length of AB.

10 mm (corresponding sides of congruent triangles equal)

13 An isosceles triangle is cut as shown, using the midpoint of AB.



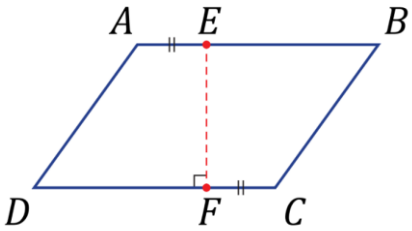
a Name the two triangles formed.

$\Delta ACM, \Delta BCM$

b Will the two triangles be congruent? Give reasons.

Yes, by SSS

14 If this parallelogram is cut as shown, explain why the two pieces are congruent.
(hint: what are the properties of a parallelogram?)



$AE = FC$ (given)

$AD = BC$ (opposite sides of parallelogram equal)

$EB = FD$ (opposite sides of parallelogram equal)

EF is common

So, all corresponding sides are equal

$\angle ADF = \angle CBE$ (opposite angles of parallelogram equal)

$\angle DAE = \angle BCF$ (opposite angles of parallelogram equal)

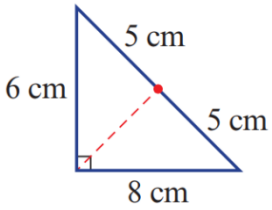
$\angle EFC = 90°$ (angles on a straight line sum to 180)

$\angle FEA = \angle EFD = 90°$ (co-interior angles between parallel lines sum to 180)

$\angle FEB = \angle EFC = 90°$ (co-interior angles between parallel lines sum to 180)

All four sides equal and all angles equal, therefore congruent.

15 If this triangle is cut as shown, determine whether the two triangles will be congruent. Explain.



No, let the red line have length x

the sides of one triangle are 5, x, 6 and the other is 5, x, 8

So they cannot be congruent as the sides are not equal.